

Mallika Dasgupta

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EDUCATION

University of California, San Diego

B.S. in Computer Science

San Diego, CA

Sep 2023 – Jun 2027

Relevant Coursework: Software Engineering, Data Science in Practice, Machine Learning : Learning Algorithms, Advanced Data Structures, Design and Analysis of Algorithms, Computer Organization, Statistical Methods, Introduction to CS Research, Systems Programming and Software Tools

EXPERIENCE

Xpertifi Skills Tech

August – September 2025

Software Engineering Intern

Mumbai, India

- Built *Xpertifi Tutor*, a RAG-based AI chatbot product that enables learners to query in-depth course material in multiple languages. The product has since been deployed by multiple financial services firms for their employee training programs.
- Developed a retrieval-augmented generation (RAG) backend using FastAPI, FAISS, and AWS Bedrock (Titan embeddings + Claude), with integrated guardrails for context-grounded conversational Q&A.
- Deployed the system on AWS (EC2 + S3) and integrated it with the Xpertifi website, where it supports nine training modules and serves over 30,000 users.

Nucsoft Ltd

July – September 2024

Software Engineering/ML Intern

Mumbai, India

- Built a machine-learning system for loan underwriting using XGBoost, Random Forest, and Linear Regression to predict loan eligibility with ~70% accuracy. Developed a Flask backend for real-time model retraining and prediction, and a React.js interface for model evaluation and loan approvals.
- Implemented clustering algorithms for a vector-based property deduplication system to improve asset verification accuracy by matching entries through name, location, and area similarity. Developed Node.js and Python APIs to enable real-time deduplication across large datasets, reducing redundancy and improving data reliability.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C, C++, HTML/CSS

Frameworks: React.js, Node.js, Flask, FastAPI

Developer Tools: Git, GitHub, Google Cloud, AWS (EC2, S3, Bedrock)

Libraries: PyTorch, scikit-learn, pandas, NumPy, Keras, Librosa, FAISS, Qiskit, Spotipy, Matplotlib, seaborn

PROJECTS

Scamurai (DiamondHacks 2025 Winner) | JavaScript, Flask, React, Gemini API

April 2025

- Participated in DiamondHacks 2025 (MLH) as part of a three-person team to build *Scamurai*, a Chrome extension that detects online scams and teaches users through interactive training.
- Built the extension using JavaScript and Flask, integrating the Gemini API to analyze messages from Gmail and WhatsApp Web for scam detection and to generate realistic, AI-driven scam scenarios for *Scamurai Dojo*—an interactive inbox simulator where users practice identifying scams in a safe environment.
- Our team was awarded **Best Overall Hack** among 300+ participants.

Triton Consulting Group- KlonIT Project (Client) | Python, Flask, Google Calendar API, MCP Protocol

April – June 2025

- Developed a Flask-based MCP server integrated with Google Calendar to schedule and manage appointments, supporting full event CRUD functionality and time-based filtering.
- Created a Python test script to simulate realistic API usage, validating response correctness, edge case handling, and MCP protocol compliance.

Triton Consulting Group | JavaScript, React.js, Node.js, MongoDB, Google OAuth

January 2025

- Led development of a MERN-stack recruitment platform to replace Google Forms and Sheets for Triton Consulting Group's application process.
- Implemented Google OAuth authentication with role-based access for applicants, associates, and administrators.
- The platform was launched for the following recruitment cycle and used by over 300 applicants to submit and track their applications.

CSes Innovate (ORCA Project) | Python, PyTorch, Librosa, FluidSynth, NumPy

November 2024 – June 2025

- Contributed to the early development of *ORCA (Orchestral Research & Computational Arrangement)*, an AI music pipeline for automated arrangement, instrument labeling, and real-time audio transcription.
- Developed modules for converting audio to MIDI, applying instrument transposition, and re-synthesizing the output using differentiable rendering in PyTorch.
- Presented the project to over 100 students and industry professionals at the Student AI Research Symposium (SAIRS).

OTHER ACTIVITIES

Swimming

- National Gold Medalist in Swimming, India.
- Member of the UCSD Club Swim Team; qualified and participated twice at the USA College Club Swimming (CCS) Nationals.